Calibration, Scoring Rules, and Conformal Prediction: Problem Set

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Contents

1 Part I – Calibration and Scoring Rules

A: Conceptual Questions

A1. Proper vs. Improper Scoring Rules

Learning objective

Understand the definition and motivation of proper scoring rules.

Question. Define a *proper* scoring rule. Give an intuitive explanation of why NLL (negative log-likelihood) is proper, while simply rewarding a classifier for outputting a high probability on the correct class regardless of magnitude would *not* be proper.

Submit: A definition and two-paragraph intuitive argument.

A2. NLL vs. Brier Score: Sensitivity Comparison

Learning objective

Compare tail-sensitivity properties of proper scoring rules.

Question. Both NLL and the Brier score are proper scoring rules for classification. Explain the key qualitative difference in how they penalise overconfident predictions. For which type of application would you prefer the Brier score, and for which would you prefer the NLL?

Submit: A qualitative comparison and two concrete application recommendations.

A3. Calibration vs. Sharpness

Learning objective

Distinguish calibration from sharpness and understand when each matters.

Question. Distinguish *calibration* from *sharpness* in probabilistic forecasting. Can a model be perfectly calibrated yet still be a poor forecaster? Can a model be sharp yet poorly calibrated? Give a brief example for each case.

Submit: Definitions of both concepts and two illustrative examples.

A4. Expected Calibration Error (ECE): Definition and Limitations

Learning objective

Know the ECE formula and its key shortcomings.

Question.

(a) Write the formula for ECE with B equal-width bins and explain each term.

(b) List two known limitations of ECE as a calibration diagnostic.

(c) What alternatives exist to address these limitations?

Submit: Formula with term-by-term explanation, two limitations, and at least two alternatives.

A5. Why Modern Deep Networks Are Miscalibrated

Learning objective

Identify architectural and training factors that cause overconfidence.

Question. According to Guo et al. (ICML 2017), modern deep networks are often accurate but *overconfident*. Describe two architectural or training factors that contribute to this miscalibration, and explain intuitively why they cause overconfidence.

Submit: Two factors with intuitive justifications.

A6. Temperature Scaling: Mechanism and Properties

Learning objective

Understand how temperature scaling modifies softmax outputs.

Question.

(a) Write the temperature-scaled softmax for logit vector $\mathbf{z} \in \mathbb{R}^C$ and scalar $T > 0$.

(b) What happens to the output when $T > 1$? When $T < 1$?

(c) Why does temperature scaling *not* change the predicted class?

(d) What is the only free parameter, and on which data split is it optimised?

Submit: Formula and short answers for each part.

B: Mathematical Derivations

B1. Strict Propriety of NLL

Learning objective

Prove that NLL is strictly proper by calculus.

Problem. Let $y \in \{0, 1\}$ with true probability $q = p^*(y = 1)$ and let the forecaster report $p \in (0, 1)$. Define the expected NLL

$$L(p) = -\big[q \log p + (1 - q) \log(1 - p)\big].$$

(a) Show that $L'(p) = 0$ if and only if $p = q$.

(b) Show that $L''(p) > 0$ for all $p \in (0, 1)$, confirming the minimum is unique.

(c) Interpret the result: what does strict propriety guarantee about the forecaster's incentive?

Submit: Full derivative calculations and one-paragraph interpretation.

B2. Strict Propriety of the Brier Score (Binary)

Learning objective

Prove strict propriety of the Brier score and interpret the minimum value.

Problem. For $y \in \{0, 1\}$ with true probability q, the expected Brier score when reporting p is

$$B(p) = q(p - 1)^2 + (1 - q)p^2.$$

(a) Expand $B(p)$ and find the value of p that minimises $B(p)$.

(b) Show that $B(q) = q(1 - q)$, and interpret this value.

Submit: Algebraic expansion and interpretation of $B(q)$.

B3. ECE Decomposition

Learning objective

Understand ECE as a weighted mean of per-bin calibration gaps.

Problem. Suppose we have N test samples and partition them into B equal-width confidence bins. Let $n_b = |\mathcal{B}_b|$, $a_b = \mathrm{acc}(\mathcal{B}_b)$, and $c_b = \mathrm{conf}(\mathcal{B}_b)$.

(a) Write the ECE formula and show it equals a weighted mean of per-bin calibration gaps.

(b) Consider $B = 1$ (one bin). What does ECE measure in this case?

(c) Suppose a model is perfectly accurate ($a_b = 1$ for all b) but confidence is not always 1. What can you say about ECE?

Submit: Formula, two-sentence answer for (b), and brief argument for (c).

B4. Temperature Scaling as a 1-D Optimisation

Learning objective

Derive the optimality condition for temperature scaling and establish convexity.

Problem. Let $\{(\mathbf{z}_i, y_i)\}_{i=1}^{M}$ be a calibration set of logits and true labels. Temperature scaling minimises

$$\mathcal{L}(T) = -\frac{1}{M}\sum_{i=1}^{M} \log \hat{q}_{y_i}(\mathbf{z}_i; T), \quad \hat{q}_c = \frac{\exp(z_{ic}/T)}{\sum_{c'} \exp(z_{ic'}/T)}.$$

(a) Let $s = 1/T$. Rewrite $\mathcal{L}$ as a function of s.

(b) Compute $\partial\mathcal{L}/\partial s$ and state the condition for optimality.

(c) Explain why this optimisation is convex in $\log T$.

Submit: Full derivation of the gradient and convexity argument.

B5. NLL as Entropy plus KL Divergence

Learning objective

Derive the decomposition $\mathbb{E}_{p^*}[-\log p_\theta] = H(p^*) + \mathrm{KL}(p^* \| p_\theta)$ from first principles, and use it to explain why minimising NLL is equivalent to minimising KL divergence and why the log score is strictly proper.

Background. In the lecture ("Link to likelihood / KL / Gibbs inequality") we stated without full proof:

$$\mathbb{E}_{p^*}\big[-\log p_\theta\big] \;=\; H(p^*) \;+\; \mathrm{KL}(p^* \,\|\, p_\theta).$$

This problem asks you to derive that result carefully and draw its consequences.

Setup. Let $\mathcal{Y}$ be a finite label set. Let p^* denote the true distribution over $\mathcal{Y}$ and p_θ a model distribution over $\mathcal{Y}$. All sums are over $y \in \mathcal{Y}$ and all logarithms are natural logarithms.

(a) **Definitions.** Write the explicit formula for each of the following, then state whether each quantity depends on θ or not:

  (i) Entropy of p^*: $H(p^*)$

  (ii) KL divergence from p_θ to p^*: $\mathrm{KL}(p^* \| p_\theta)$

  (iii) Cross-entropy of p_θ relative to p^*: $H(p^*, p_\theta)$

  (iv) Expected NLL when data $y \sim p^*$: $\mathcal{L}(\theta) = \mathbb{E}_{y\sim p^*}[-\log p_\theta(y)]$

(b) **Derivation.** Starting *only* from $\mathcal{L}(\theta) = -\sum_y p^*(y) \log p_\theta(y)$, derive the decomposition

$$\mathcal{L}(\theta) = H(p^*) + \mathrm{KL}(p^* \| p_\theta)$$

in no more than four clearly labelled algebraic steps.
Hint: add and subtract $\log p^*(y)$ *inside the sum.*

(c) **Gibbs' inequality.** State Gibbs' inequality for $\mathrm{KL}(p^* \| p_\theta)$ and use it to prove that:

  (i) $\mathcal{L}(\theta) \geq H(p^*)$, with equality iff $p_\theta = p^*$.

  (ii) Minimising the expected NLL over θ is equivalent to minimising $\mathrm{KL}(p^* \| p_\theta)$ (since $H(p^*)$ does not depend on θ).

(d) **Link to maximum likelihood.** Explain in two sentences why minimising the *average* NLL over a training set $\{y_i\}_{i=1}^N$ is equivalent to maximising the likelihood $\prod_{i=1}^N p_\theta(y_i)$.

(e) **Strict propriety.** Use the decomposition to give a one-sentence proof that the log score $S(p_\theta, y) = \log p_\theta(y)$ is *strictly proper*: i.e., the expected score is maximised *if and only if* $p_\theta = p^*$.

Submit: Explicit formulas in (a); a four-step derivation in (b) showing each algebraic manipulation; Gibbs' inequality stated and applied in (c); two sentences for (d) and (e).

C: Parametric Analysis and Small Examples

C1. Effect of Temperature on NLL and Brier Score

Learning objective

Trace how miscalibration affects multiple metrics simultaneously.

Question. Start from a perfectly calibrated model ($T = 1$, ECE ≈ 0). Qualitatively describe what happens to NLL, Brier score, and ECE as you:

(a) Increase T from 1 to a large value (e.g. $T = 5$).

(b) Decrease T from 1 toward 0 (e.g. $T = 0.2$).

Submit: Qualitative direction of change for each metric under each scenario, with brief justification.

C2. Behaviour Under Class Imbalance

Learning objective

Recognise when calibration metrics produce misleading results under imbalance.

Question. Suppose the test set is highly imbalanced: class 1 (positive) has prevalence $\pi = 0.05$. A classifier always predicts "negative" with confidence 0.95.

(a) Compute the ECE of this classifier (assume all predictions fall in one bin).

(b) Compute the expected Brier score.

(c) Compute the expected NLL.

(d) Is this classifier well-calibrated? Is it useful?

Submit: Numerical answers for (a)–(c) and a one-paragraph assessment for (d).

C3. Isotonic Regression vs. Histogram Binning under Small Calibration Sets

Learning objective

Understand the overfitting risk of flexible calibration methods with small datasets.

Question. Suppose you have only $M = 50$ samples in your calibration set. Compare histogram binning (with $K = 10$ bins) and isotonic regression in terms of:

(a) Expected number of samples per bin (for histogram binning).

(b) Overfitting risk for each method.

(c) Which method you would recommend and why.

Submit: A calculation for (a) and a comparative recommendation.

C4. Discrete Cross-Entropy = Entropy + KL: Numeric Verification (New)

Learning objective

Verify the NLL decomposition numerically on a concrete three-class example.

Problem. Consider a single input x with the following distributions over three classes $\{1, 2, 3\}$:

	Class 1	Class 2	Class 3
True distribution $p^*(y \mid x)$	0.50	0.30	0.20
Model distribution $p_\theta(y \mid x)$	0.70	0.20	0.10

(a) Compute $H(p^*) = -\sum_y p^*(y \mid x) \log p^*(y \mid x)$ (use natural logarithm; give your answer in nats to three decimal places).

(b) Compute $\mathrm{KL}(p^* \| p_\theta) = \sum_y p^*(y \mid x) \log \dfrac{p^*(y \mid x)}{p_\theta(y \mid x)}$.

(c) Compute the cross-entropy $H(p^*, p_\theta) = -\sum_y p^*(y \mid x) \log p_\theta(y \mid x)$ directly from the definition.

(d) Verify numerically that $H(p^*, p_\theta) = H(p^*) + \mathrm{KL}(p^* \| p_\theta)$.

(e) Which of $H(p^*)$ and $H(p^*, p_\theta)$ is larger? What does this tell you about whether the model is adding or reducing uncertainty?

Useful values: $\ln(0.5) = -0.693$, $\ln(0.3) = -1.204$, $\ln(0.2) = -1.609$, $\ln(0.7) = -0.357$, $\ln(0.1) = -2.303$.

Submit: Numerical computations for each part with intermediate steps shown.

D: Applied Problems and Notebook Extensions

D1. Engineering Application – Medical Diagnostic Classifier

Learning objective

Interpret calibration metrics in a safety-critical engineering context.

A binary classifier is trained to detect a cardiac condition from ECG features. It outputs a probability $\hat{p}$ of the condition being present. After training, on a held-out test set of $N = 500$ patients:

- Accuracy = 88%
- ECE = 0.14 (overconfident; high-confidence bins have accuracy ≈ 0.72)
- Mean confidence in the positive class when predicting positive = 0.91

(a) Why is a high ECE of 0.14 particularly dangerous in this medical context?

(b) You apply temperature scaling and find $T^* = 1.8$. What does $T^* > 1$ tell you about the original model's predictions?

(c) After temperature scaling ECE drops to 0.03 and accuracy remains 88%. Explain why accuracy is unchanged.

(d) A colleague argues "since we care about whether the patient has the condition or not, we should only report accuracy, not probabilities." Give two arguments against this position.

Submit: Short justifications for each part; no numerical computation needed beyond what is given.

D2. Notebook Extension: Comparing All Four Calibration Methods

Learning objective

Design a fair comparative experiment for post-hoc calibration methods.

The Guo et al. (ICML 2017) results for CIFAR-100/ResNet-110 show: Uncalibrated (ECE = 12.67), Temperature Scaling (ECE = 0.96), Histogram Binning (ECE = 2.46), Isotonic Regression (ECE = 4.16).

(a) Why does temperature scaling outperform isotonic regression on this large test set, contrary to the usual assumption that more flexible methods do better?

(b) Write a short experimental plan (3–4 sentences) for fairly comparing temperature scaling, Platt scaling, histogram binning, and isotonic regression on the `sklearn` Digits dataset.

(c) How would the relative performance of isotonic regression vs. temperature scaling change if the calibration set size grew from $M = 100$ to $M = 5000$? Justify your answer.

Submit: Prose answers for (a) and (c); a concise experimental plan for (b).

2 Part II – Conformal Prediction

Suggested pre-reading. Before attempting Part II, review the following:

- Angelopoulos & Bates (2023), 'Conformal Prediction: : A Gentle Introduction," *Foundations and Trends in ML*.
- Shafer & Vovk (2008), "A Tutorial on Conformal Prediction," *JMLR* 9:371–421.

A: Conceptual Questions

A1. What Conformal Prediction Guarantees (and Does Not)

Learning objective

State and interpret the marginal coverage guarantee precisely.

Question.

(a) State the marginal coverage guarantee of split conformal prediction in one sentence, clearly identifying what probability is lower-bounded.

(b) Explain the difference between *marginal* and *conditional* coverage. Give a concrete example where marginal coverage is satisfied but conditional coverage fails.

(c) What single assumption is needed for the guarantee to hold? Why is this assumption weaker than assuming a specific data distribution?

Submit: A precise one-sentence guarantee, a definition of each coverage type, a concrete counterexample, and a brief statement of the assumption.

A2. The Role of the Nonconformity Score

Learning objective

Compute and interpret two nonconformity scores used in classification.

Question. Two nonconformity scores are commonly used for classification:

- **Threshold score:** $s_{\text{th}}(x, y) = 1 - p(y \mid x)$
- **APS score:** $s_{\text{APS}}(x, y) = \sum_{y' : p(y'|x) \geq p(y|x)} p(y' \mid x)$

(a) Explain intuitively what each score measures. Why does a larger score mean "less conforming"?

(b) For a classifier with softmax output (0.70, 0.20, 0.10) on classes (A, B, C), compute the APS score for the true label $y = B$.

(c) Under what conditions would you prefer the APS score over the threshold score?

Submit: Intuitive explanation, numerical answer for (b), and a preference justification.

A3. Conformal Prediction vs. Bayesian Credible Intervals

Learning objective

Contrast frequentist conformal guarantees with Bayesian credible intervals.

Question.

(a) What assumption does the Bayesian interval require that the conformal interval does not?

(b) In what sense is the conformal coverage guarantee "stronger" than a Bayesian credible interval's guarantee?

(c) Can conformal prediction be applied on top of a Bayesian model? If so, what does it add?

Submit: One paragraph per part.

A4. Coverage vs. Efficiency Trade-off

Learning objective

Understand why trivial predictors are uninformative and define efficiency.

Question.

(a) Explain why returning $\mathcal{T}(x) = \mathcal{Y}$ (the full label set) always achieves 100% coverage but is useless.

(b) Define an *efficient* conformal predictor. What is the ideal relationship between coverage and average set size?

(c) How does model quality affect efficiency? Would a random classifier produce small or large average set sizes at the same coverage level?

Submit: Concise answers with brief justifications.

A5. Distribution Shift and Conformal Validity

Learning objective

Explain how and why the coverage guarantee breaks under distribution shift.

Question.

(a) Explain why the coverage guarantee fails under distribution shift.

(b) You calibrate on clean Digits images and test on images with added Gaussian noise. Would you expect empirical coverage to be above or below $1 - \alpha$? Justify.

(c) Propose one strategy to recover the coverage guarantee when distribution shift is known to occur.

Submit: A mechanistic explanation for (a), a directional prediction for (b) with justification, and one concrete strategy for (c).

Question.

(a) List and briefly explain *four* criteria or practices for selecting a valid held-out calibration dataset for conformal prediction (consider: exchangeability, data leakage, temporal ordering, group structure, stratification, and size).

(b) Describe concretely what can go wrong if calibration data are *not* exchangeable with test data. Give a specific example from an engineering or medical setting.

(c) How should you choose the miscoverage level α, and what does $1-\alpha$ mean in practice for a decision-maker? Give one example where $\alpha = 0.10$ is appropriate and one where it is too lax.

(d) A practitioner says: "I will use my entire labelled dataset for both calibration and testing to maximise efficiency." What is wrong with this approach?

Submit: Concise answers for each part; (a) should list four distinct criteria.

B: Mathematical Derivations

B1. The Finite-Sample Coverage Proof

Learning objective

Prove the conformal coverage guarantee using a rank argument.

Problem. Let $s_1, \ldots, s_n$ be i.i.d. nonconformity scores on a calibration set, and let s_{n+1} be the score of a new test point, with all $n+1$ scores i.i.d. Let $\hat{q} = s_{(\lceil (1-\alpha)(n+1) \rceil)}$, the k-th smallest among $s_1, \ldots, s_n$ where $k = \lceil (1-\alpha)(n+1) \rceil$.

(a) Under exchangeability, what is the distribution of the rank of s_{n+1} among $\{s_1, \ldots, s_n, s_{n+1}\}$?

(b) Using the rank argument, show that $\Pr(s_{n+1} \le \hat{q}) \ge 1-\alpha$.

(c) What is the exact upper bound on $\Pr(s_{n+1} \le \hat{q})$? What does this say about "conservative" conformal predictors?

Submit: A full proof with the rank argument and explicit probability bounds.

B2. Split Conformal Regression: Interval Construction

Learning objective

Derive and analyse the conformal prediction interval for regression.

Problem. A regression model $\hat{y}(x)$ is trained on a training set. On a calibration set of size n, absolute residuals $s_i = |y_i - \hat{y}(x_i)|$ are computed. The conformal prediction interval is $C(x) = [\hat{y}(x) - \hat{q},\ \hat{y}(x) + \hat{q}]$, where $\hat{q} = \text{Quantile}_{1-\alpha}(\{s_i\}_{i=1}^{n})$.

(a) Show that $C(x)$ has the form derived from the nonconformity score $s(x, y) = |y - \hat{y}(x)|$.

(b) Suppose $n = 99$ calibration points and $\alpha = 0.10$. What is the exact order statistic used for $\hat{q}$ (using the $(n+1)$ correction)?

(c) The interval $C(x)$ has *constant width* $2\hat{q}$ regardless of x. Give one practical scenario where this is problematic, and name a method that addresses it.

Submit: Derivation for (a), numerical answer for (b), and a scenario with method name for (c).

B3. Effect of Calibration Set Size on the Quantile

Learning objective

Analyse the large-sample and finite-sample behaviour of the conformal quantile.

Problem. Let $s_1, \ldots, s_n \overset{\text{i.i.d.}}{\sim} F$ for some continuous CDF F. The conformal quantile is $\hat{q} = s_{(\lceil (1-\alpha)(n+1) \rceil)}$.

(a) As $n \to \infty$, what does $\hat{q}$ converge to?

(b) For finite n, is $\hat{q}$ a biased or unbiased estimate of $F^{-1}(1-\alpha)$? In which direction?

(c) Compute the minimum calibration size n required so that the finite-sample overcoverage $1/(n+1)$ is at most 0.005.

Submit: Asymptotic result for (a), bias direction with justification for (b), and exact computation for (c).

B4. APS Score: Computation and Coverage

Learning objective

Compute APS scores for all labels and form a prediction set.

Problem. Consider a 4-class classifier with softmax output $\hat{\mathbf{p}} = (0.50, 0.30, 0.15, 0.05)$ for classes (A, B, C, D).

(a) Compute the APS nonconformity score for each possible true label $y \in \{A, B, C, D\}$.

(b) Suppose the calibration threshold is $\hat{q} = 0.85$. Which classes are included in the prediction set $\mathcal{T}(x)$?

(c) If $\hat{q} = 0.85$ achieves 90% coverage, interpret what this means for 1000 test inputs.

Submit: Table of APS scores, prediction set, and interpretation.

C: Parametric Analysis

C1. Effect of α on Set Size and Coverage

Learning objective

Trace how the miscoverage level controls the coverage–efficiency trade-off.

Question.

(a) As α increases from 0 to 0.30, how does $\hat{q}$ change, and why does the average set size decrease?

(b) What property of split conformal prediction ensures empirical coverage stays near or above $1-\alpha$ for all tested α?

(c) In the notebook, Threshold produces smaller sets than APS but may under-cover at larger α. Explain this apparent contradiction.

Submit: Concise answers with brief justifications.

C2. Mondrian (Class-Conditional) Conformal Prediction

Learning objective

Understand per-class calibration and its coverage guarantee.

Question.

(a) What coverage guarantee does Mondrian conformal provide, and how does it differ from marginal coverage?

(b) Suppose class $c = 1$ has 200 calibration points and class $c = 2$ has only 20. Compare the reliability of $\hat{q}_1$ vs. $\hat{q}_2$ for $\alpha = 0.10$. How many samples per class are needed to limit overcoverage to 0.005?

(c) Give one real-world scenario where Mondrian conformal prediction is substantially better than standard split conformal.

Submit: Guarantee statement, numerical comparison, and one concrete scenario.

C3. Conformalized Quantile Regression (CQR) vs. Standard Conformal

Learning objective

Derive the CQR prediction interval and understand its adaptive-width property.

Question. Standard conformal regression uses $s(x, y) = |y - \hat{y}(x)|$, producing constant-width intervals. CQR uses conditional quantile predictions $\hat{q}_\alpha^-(x)$ and $\hat{q}_\alpha^+(x)$ with score:

$$s(x, y) = \max\bigl(\hat{q}_\alpha^-(x) - y,\ y - \hat{q}_\alpha^+(x)\bigr).$$

(a) Interpret the CQR score geometrically: what does a positive score mean?

(b) Show that the CQR prediction interval takes the form $[\hat{q}_\alpha^-(x) - \hat{q},\ \hat{q}_\alpha^+(x) + \hat{q}]$.

(c) The CQR interval is wider in the high-x region than the low-x region. What does this indicate about the data-generating process?

Submit: Geometric interpretation, algebraic derivation for (b), and data interpretation for (c).

D: Applied Problems and Notebook Extensions

D1. Engineering Application – Structural Load Prediction

Learning objective

Apply split conformal regression in an engineering context and reason about exchangeability violations.

A civil engineer uses a regression model $\hat{y}(x)$ to predict the maximum load (kN) a structural component can bear. She calibrates on $n = 199$ components at $\alpha = 0.05$, obtaining $\hat{q} = 8.3\,\text{kN}$.

(a) Interpret $C(x) = [\hat{y}(x) - 8.3,\ \hat{y}(x) + 8.3]$ in plain engineering language.

(b) The engineer argues that a constant-width interval is unsafe because some components are more variable. What method would you recommend?

(c) A design change shifts the distribution of x (new alloy). Without recalibrating, would you expect over- or under-coverage? Justify.

(d) How many new components must be tested to recalibrate and achieve at most 0.5% overcoverage at $\alpha = 0.05$?

Submit: Plain-language interpretation, method recommendation, coverage direction with justification, and a numerical minimum-n calculation.

D2. Notebook Analysis: Classification Results

Learning objective

Interpret conformal classification experiment results and implement Mondrian conformal.

The notebook (`conformal_prediction_lab_notebook.ipynb`) trains a classifier on the Digits dataset and plots coverage and average set size vs. α.

(a) Both Threshold and APS coverage curves lie *above* $1 - \alpha$ for all tested α. Is this expected? Could either curve ever dip below the nominal line?

(b) APS produces larger average sets than Threshold for all $\alpha > 0.05$, yet APS is described as "often more efficient." Explain this apparent contradiction.

(c) Propose code to test Mondrian conformal prediction. Write key Python pseudocode for the per-class calibration step.

(d) If the classifier is replaced by a random (uniform) one, what would the efficiency plot look like? Would coverage still hold?

Submit: Prose answers for (a) and (b); Python pseudocode for (c); qualitative prediction for (d).

D3. Notebook Extension: Regression and Distribution Shift

Learning objective

Design and analyse conformal regression experiments including distribution shift.

The notebook demonstrates split conformal regression and CQR on a synthetic dataset, reporting: Split Conformal coverage = 89.4%, width = 8.08; CQR coverage = 85.6%, avg. width = 8.34 at nominal 90%.

(a) Give two possible explanations for the slight under-coverage.

(b) For the **shift test** (calibrate on clean Digits, test on noisy Digits), write a short experimental plan: what metric, how to generate the shift, and how to detect when the guarantee has broken.

(c) Describe how to define "difficulty buckets" for a Mondrian variant on the regression task, and write pseudocode for the bucketed calibration step.

Submit: Two explanations, a 3–4 sentence experimental plan, and annotated pseudocode.

E: Analytical Table Problems

E1. Fill-the-Table: Classification Conformal Prediction

Learning objective

Compute Threshold and APS scores by hand, find the conformal quantile, and form prediction sets for new test points.

Setup. A 3-class image classifier outputs softmax probabilities over classes $\{A, B, C\}$. The calibration set has $n = 5$ points with the predicted probability vectors and true labels shown below:

i	$p(A \mid x_i)$	$p(B \mid x_i)$	$p(C \mid x_i)$	True y_i	$s_{\text{th},i} = 1 - p(y_i \mid x_i)$
1	0.70	0.20	0.10	A	
2	0.50	0.30	0.20	B	
3	0.15	0.65	0.20	B	
4	0.10	0.20	0.70	C	
5	0.40	0.35	0.25	A	

Use $\alpha = 0.20$ throughout.

(i) **Threshold score.** Fill in column $s_{\text{th},i} = 1 - p(y_i \mid x_i)$ for all five calibration points.

(ii) **APS score.** For each calibration point i, compute $s_{\text{APS},i} = \sum_{y' : p(y'|x_i) \geq p(y_i|x_i)} p(y' \mid x_i)$. Show which classes are summed for each row. *Tie-handling:* when two classes have equal probability, include *both*.

(iii) **Quantile index.** Compute $k = \lceil (n+1)(1-\alpha) \rceil$. Show the calculation.

(iv) $\hat{q}$ **for each method.** Sort the Threshold scores and sort the APS scores separately, then read off $\hat{q}_{\text{th}} = s_{(k)}$ and $\hat{q}_{\text{APS}} = s_{(k)}$.

(v) **Prediction sets for Test Point 1.** Given $\hat{\mathbf{p}}_{\text{test1}} = (0.55, 0.30, 0.15)$:

- Form $\mathcal{T}_{\text{th}}(x)$ using the rule $1 - p(y \mid x) \leq \hat{q}_{\text{th}}$.

- Compute all three APS scores for this test point and form $\mathcal{T}_{\text{APS}}(x)$ using $s_{\text{APS}}(x, y) \leq \hat{q}_{\text{APS}}$.

(vi) **Prediction sets for Test Point 2.** Given $\hat{\mathbf{p}}_{\text{test2}} = (0.28, 0.27, 0.45)$: Repeat the above for both methods.

(vii) **Comparison.** Fill in the table below and explain in one sentence why the two methods give different set sizes.

	Threshold set	APS set
Test 1		
Test 2		

Submit: Completed tables, all intermediate computations shown, and the one-sentence comparison.

E2. Fill-the-Table: Regression Conformal Prediction

Learning objective

Apply the split conformal regression algorithm by hand: compute residual scores, find the quantile, and construct prediction intervals.

Setup. A regression model $\hat{y}(x)$ predicts a continuous quantity. The calibration set has $n = 5$ points:

i	x_i	$\hat{y}(x_i)$	y_i (observed)	$s_i = \lvert y_i - \hat{y}(x_i)\rvert$
1	1.0	2.5	2.8	
2	2.0	4.0	3.3	
3	3.0	5.5	6.1	
4	4.0	7.0	5.8	
5	5.0	8.5	9.8	

Use $\alpha = 0.20$ throughout.

(i) **Residual scores.** Fill in the column $s_i = |y_i - \hat{y}(x_i)|$ for all five calibration points.

(ii) **Quantile index.** Compute $k = \lceil (n+1)(1-\alpha) \rceil$ and identify $\hat{q} = s_{(k)}$ (the k-th order statistic of the calibration scores).

(iii) **Prediction intervals.** For each test point below, construct the interval $C(x) = [\hat{y}(x) - \hat{q},\ \hat{y}(x) + \hat{q}]$.

Test point	$\hat{y}(x)$	$C(x) = [\hat{y}(x) \pm \hat{q}]$
$x = 2.5$	4.75	
$x = 4.5$	7.75	

(iv) **Edge case.** Repeat the computation of k and $\hat{q}$ with $\alpha = 0.10$. What happens? Explain what this means in practice.

(v) **Width interpretation.** Do both test-point intervals have the same width? Is this a feature or a limitation of standard conformal regression? Name one method that produces *adaptive-width* intervals.

Submit: Completed tables, intermediate computations, edge-case discussion, and a one-sentence answer to (v).

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